

Jimmy Dagum, Jr.

702-978-2420 | jimmydagumjr@gmail.com | jimmydagumjr.com

Education

University of Nevada, Las Vegas

Bachelor of Science in Computer Science

Las Vegas, NV

Expected Graduation May 2026

Technical Skills

Languages: C++, C, Python, TypeScript, JavaScript, Go, PHP, HTML, CSS, SQL, x86-64 ASM/MIPS, MATLAB

Frameworks & Libraries: Next.js, React, Zustand, TanStack Query, Prisma, Tailwind CSS, Socket.IO, Bootstrap, Framer Motion

Databases: SQL Server, MySQL, PostgreSQL

Tools / Platforms: Git, GitLab CI/CD, Bash, Proxmox, RabbitMQ (AMQP), SSH, SSMS, Quest KACE, WebSockets, SMTP, Docker

Experience

Software Engineer Intern

Nov 2025 – Present

University of Nevada, Las Vegas

Las Vegas, NV (Remote)

- Modernized a legacy air quality modeling system by migrating a MATLAB-based SCMB tool into a multi-user Next.js and Node.js web application, enabling browser-based execution
- Standardized production deployments by containerizing the MATLAB runtime and backend services with Docker for Linux deployments, reducing environment-specific failures and setup time
- Enabled repeatable experimentation workflows by building RESTful APIs for simulation execution, scenario persistence, and multi-sheet Excel result exports

Software Engineer Intern

May 2025 – Aug 2025

JT4

Las Vegas, NV

- Strengthened platform security and developer velocity by migrating raw SQL Server queries to Prisma ORM and implementing role-based authentication within a Next.js application, reducing schema errors and access misconfiguration
- Reduced database load by 90% by implementing feature-level in-memory caching and optimizing query patterns for high-traffic endpoints
- Doubled email pipeline throughput by integrating RabbitMQ with a Go microservice using batched fan-out, retries, and failure re-queuing

Student Systems Administrator

Mar 2025 – May 2025

University of Nevada, Las Vegas

Las Vegas, NV (Remote)

- Automated system maintenance workflows by developing Python scripts against the KACE API and integrating them into GitLab CI/CD pipelines, reducing manual update overhead
- Hardened infrastructure reliability by securely decommissioning 10+ legacy Linux and Windows virtual machines

Software Engineer Intern

May 2024 – Dec 2024

NexOasis – Enterprise Security

Las Vegas, NV (Remote)

- Reduced VM provisioning time by 80% by building a Proxmox-based web interface that automated certificate authority setup, VM cloning, and secure SSH file transfers
- Optimized data retrieval performance by 30% by designing MySQL schemas to map user roles to VM identifiers and cron-driven automation workflows

Information Technology Support Consultant

Dec 2023 – May 2024

University of Nevada, Las Vegas

Las Vegas, NV

- Enabled real-time campus energy visibility by developing a React dashboard that visualized real-time solar production across 16 buildings using internal APIs
- Accelerated operational reporting by automating consolidation of 10,000+ work orders using Python and JavaScript

Projects

Catharsis | React, Redux, PostgreSQL, Node.js, Git

- Enabled large-scale media hosting by building a full-stack React platform for photo galleries and music streaming, supporting thousands of users
- Scaled data storage and retrieval by designing PostgreSQL schemas and asynchronous retrieval pipelines for user libraries and metadata
- Strengthened account security and state reliability by implementing SMTP-based two-factor authentication and centralized global state management with Redux

Battery Workforce Challenge (UNLV) | MATLAB, Simulink, Git

- Boosted battery modeling accuracy by 30% by developing lithium-ion cell simulations using real hardware telemetry in MATLAB and Simulink
- Increased charge prediction accuracy by 25% by implementing SoC/SoH estimation with Kalman filtering, reducing runtime by 10 seconds